

Presented By Team Labrador (Sam, Elias and Joelle)

How are parental
education levels
associated
with
traditional
gender-role
attitudes?

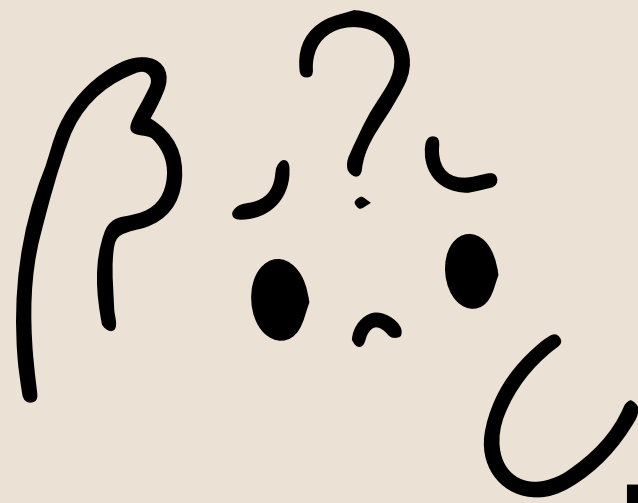


Our Hypotheses

Alternative Hypotheses:

H1a – Mother's education level is associated with traditional gender-role attitudes ($r \neq 0$)

H1b – Father's Education level is associated with traditional gender-role attitudes ($r \neq 0$)



Null Hypotheses:

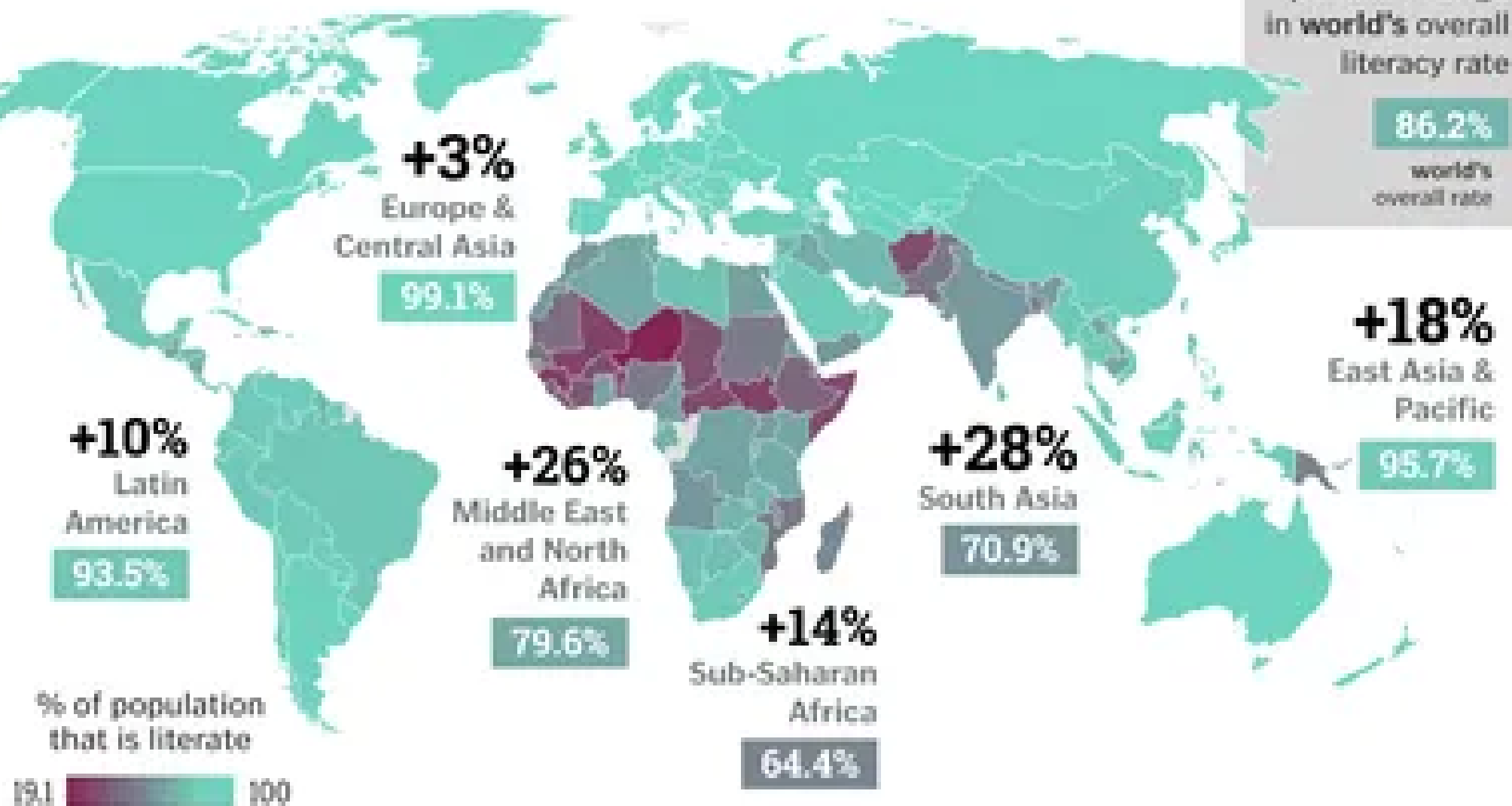
H0a – Mother's education level is not associated with traditional gender-role attitudes ($r = 0$)

H0b – Father's education level is not associated with traditional gender-role attitudes ($r = 0$)



CHANGES IN GLOBAL LITERACY OVER TIME

Increase in adult literacy rate from 1987 to 2016



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 Note: No data for the percent change in literacy rates in the US and Canada
 Source: World Bank

GZERO

		Weather	
Temperature		Forecast: Light to moderate rain at a few places with thunderstorm and lightning is likely to occur over Tamil Nadu and Puducherry	
MIN	MAX		
28°C	38°C	Air quality (PM 10) at 8 pm	
Sunrise	5:44am	Rayapetam	70 ug/m3
Sunset	6:38pm	Marol	38 ug/m3
Moonrise	12:52am	Perungudi	111 ug/m3
Moonset	1:38pm		

With over 8 crore population, Tamil Nadu set to achieve Centre's target 2 years before deadline

State to be fully literate by year end

■ NIRUPA SAMPATH

CHENNAI: Tamil Nadu is set to achieve full literacy by the end of 2025, two years ahead of the deadline set by the Union government. This will be attained after the state Directorate of Non-Formal and Adult Education (DNFE) will conduct the final round of assessment by November under the New India Literacy Programme (NILP).

NILP, under the Ministry of Education, is a five-year programme from financial years 2021-27 to educate adults. In India, however, TN, one of the largest states with over eight crore population, has achieved this feat in just three years. TN officials claim that the government had set 80% as the target to achieve full literacy.

Meanwhile, sources say that Ladakh, with over 3.02 lakh population, has already achieved full literacy with Goo (15.75 lakh population), and Mizoram (12.38 lakh population) is in line to announce full literacy in the future, but these are states with lesser population compared to TN.

A few districts like Tiruvannamalai, Salem and Dharmapuri have uneducated persons in some pockets; they will be targeted in the November FLNAT
 — S Nagaraja Marugan, Director, DNFE

Literacy is achieved when a learner above the age of seven (studying in class 1) is able to read, write and do basic arithmetic. A state/country is declared fully literate when it achieves 95% literacy.

Explaining how TN will achieve this mark, S Nagaraja Marugan, director of DNFE, under the Directorate of School Education, said, "The literacy rate of Tamil Nadu 80.89%, as per the 2011 census. However, over the span of 14 years, the state has shown outstanding performance with a few districts already breaching the 95% literacy mark. Hence, we will be able to declare full literacy by 2025-end."

"A few northern districts, like Thiruvannamalai, Salem and Dharmapuri, have uneducated persons in some concentrated regions; they will be included in the November FLNAT," noted the director.

Meanwhile, according to the data of Periodic Labour Force Survey (PLFS) 2023-24 in connection with the Sustainable Development Goals India Index 4.6, TN was performer number one under SDG 4.6 (percentage of persons 15 years and above) who are literate.

With the mission to eradicate illiteracy, DNFE under the state government has been conducting the Foundational Literacy and Numeracy Programme (FLNAT) from 2022-23 academic year for learners above 15 years across the country.

"TN, despite its population, has achieved this feat within the deadline set by both the state and union governments, which is 2026 and 2027, respectively. Among the districts, Ramnathapuram has already achieved full literacy, with 236 learners recently clearing the FLNAT," added the director.

And, as per DNFE data, from 2022-23 till June 2025, 30.56 lakh adults in the age group of 15 years and above have been benefited and provided basic literacy under NILP. Of the 30.56 lakh, the majority are women (16.15 lakh) who had appeared and cleared FLNAT.



Elderly men and women attend a literacy test at an NILP centre in Madurai

THE ROAD TO 95%

The literacy rate of Tamil Nadu was 80.89% as per the 2011 census.

Over 14 years, the state has shown outstanding performance with a few districts breaching the 95% literacy mark.

Hence, the state is set to achieve full literacy by 2025-end.

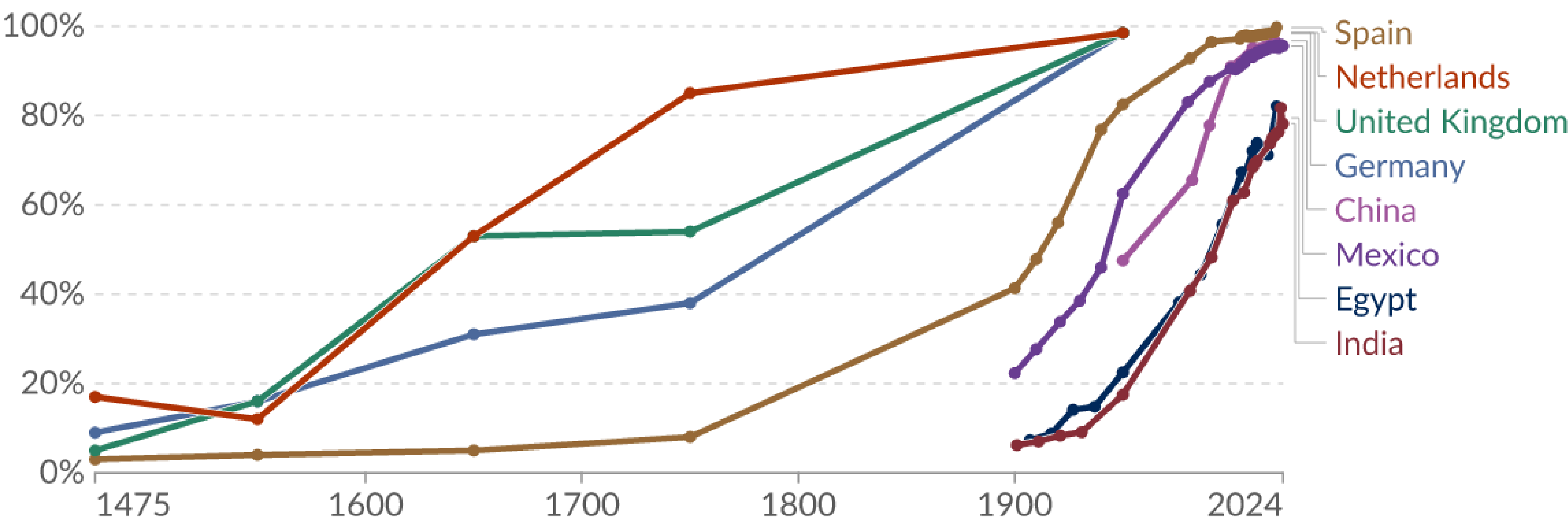
What is literacy?

Literacy is achieved when a learner above the age of seven is able to read, write and do basic arithmetic. A state/country is declared fully literate when it achieves 95% literacy.



Literacy rate 1475 to 2024

Share of adults who can read and write a simple statement about their everyday life.



Data source: UNESCO (1957) and other sources

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Note: Because definitions of literacy and population coverage changed over time, comparisons between historical and modern data should be made with caution.

Gender equality woefully off track, new UN report reveals



The debate on traditional gender roles has taken a turn

A new study suggests that a larger proportion of younger men in Singapore hold traditional views of masculinity than older generations. CNA TODAY looks into what is shaping these beliefs and what this means for men and women alike.

123 years. That's how long it will take to reach gender parity, according to the World Economic Forum's latest [Global Gender Gap Report](#).

It's a long time, but in 2024 that figure was 132 years, so we've come 11 years closer to full gender parity in the past 12 months.



Independent variable:

**Parental Education Levels
(Split between Mothers and Fathers)**

Dependent variable:

Traditional Gender-role attitudes

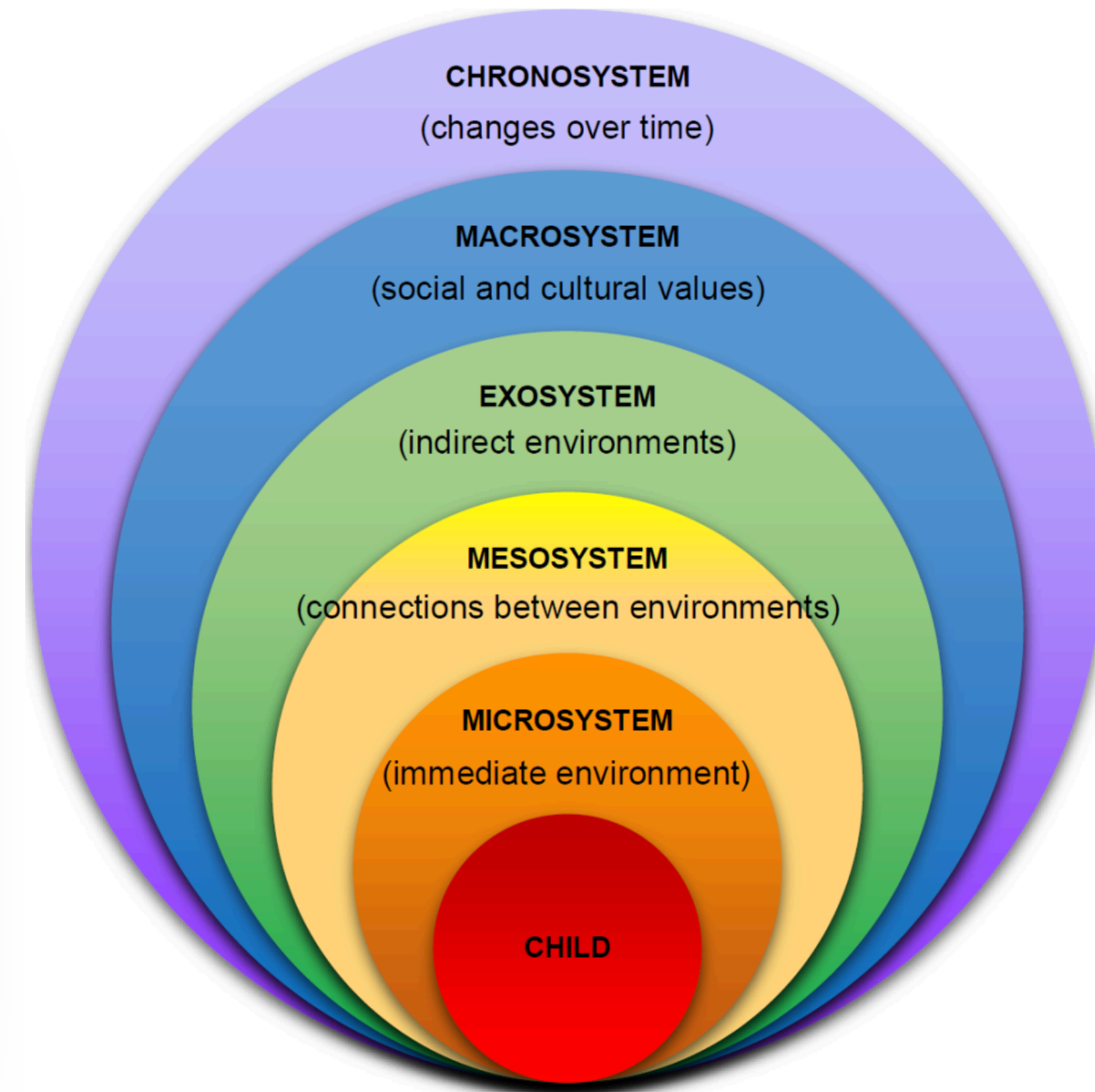
Theory our Research is based on:

Socialisation Theory

Socialisation Theory

- Socialization” as the lifelong process of inheriting and disseminating norms, customs, and ideologies, providing an individual with the skills and habits necessary for participating within his or her own society.
- “the means by which social and cultural continuity are attained.”

Key Agents of Socialisation:
Peer Groups, Media, **Family**...

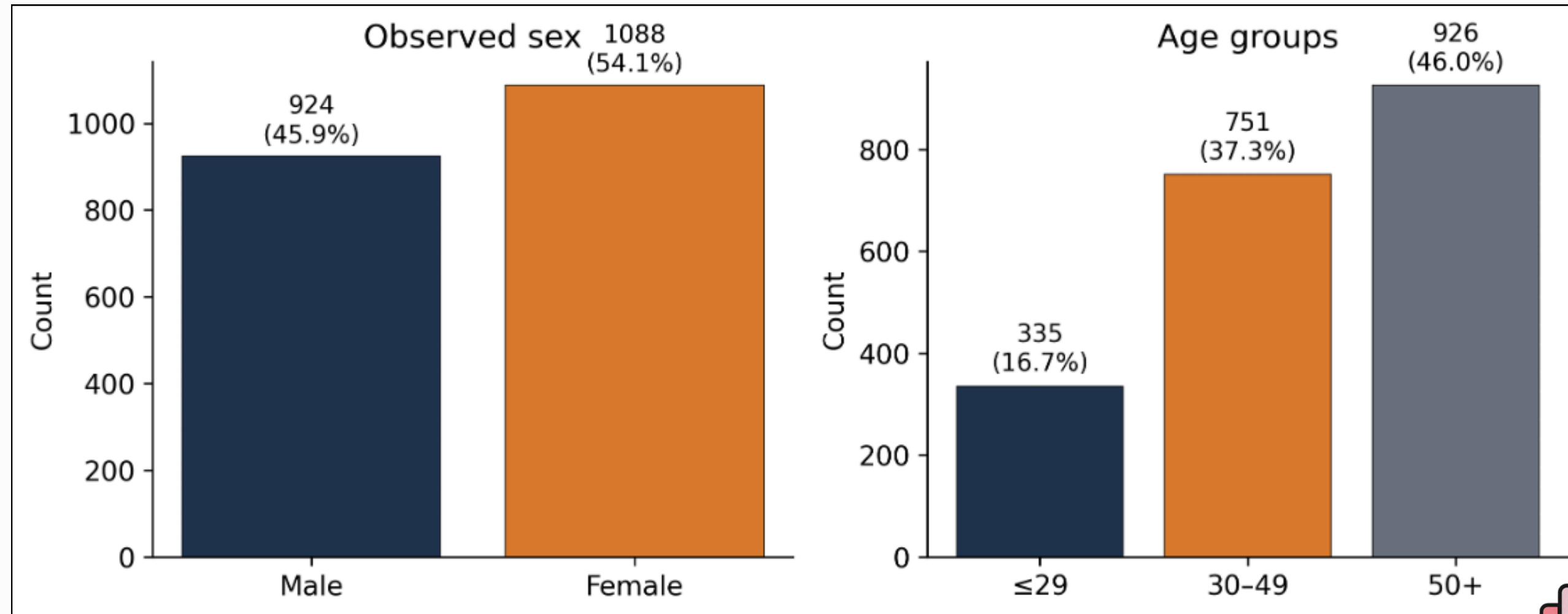


Why did we choose Singapore?

- **Rapid transformation:** Industrialisation created new employment opportunities for women.
- **Rising participation:** Female labour-force participation rose from 29.5% (1970) → 53% (1990); working women more than doubled in 1970–80.
- **Education:** Improved female education expanded career opportunities and supported women's economic participation.
- **Changing gender roles:** The rise of the "working woman" challenged traditional gender divisions of labour.

- **Education & transformation:** Improved female education expanded career opportunities and supported women's economic participation.
- **Why Singapore?** Its educational transformation provides a relevant context for studying parental education and gender-role attitudes.
- **Individual-level analysis:** WVS allows us to examine this relationship at the individual level, beyond national employment trends.
- **Shared context:** Focusing on Singapore controls for broad institutional, educational and social differences between countries.

Who took the study?



The Singapore WVS sample consisted of 2,012 respondents residing in Singapore; 90.4% were Singapore citizens



Cleaning the Data Set



Data Cleaning Process

1. Select relevant variables

- Extracted 15 WVS variables relevant to our research questions
- Gender roles (Q27–Q32), science/religion (Q158–Q164), and parental education (Q277R–Q278R)

2. Handle missing values

- Recoded WVS missing-value codes (–1 to –5) as NaN
- Retained respondents with valid data rather than deleting entire cases

3. Check and align coding

- Reviewed question wording and response scales
- Reverse-coded Q160–Q162 so higher scores consistently indicate more positive attitudes toward science & technology
- Confirmed parental education is ordered from lower → middle → higher

4. Save cleaned dataset

- Exported the cleaned variables as WVS_Singapore_cleaned.csv
- This dataset forms the basis for subsequent IV and DV construction and analysis



Creating our Traditional Gender-role Attitudes Score

Reverse-Coding

Cronbach's Alpha

Traditional Gender Role
Attitude Score

- Used Q28 – Q31 as measures of Traditional Gender-role Attitudes

For each of the following statements I read out, can you tell me how strongly you agree or disagree with each. Do you strongly agree, agree, disagree, or strongly disagree? (Read out and code one answer for each statement):

		Strongly agree	Agree	Disagree	Strongly disagree
Q27	One of my main goals in life has been to make my parents proud	1	2	3	4
Q28	When a mother works for pay, the children suffer	1	2	3	4
Q29	On the whole, men make better political leaders than women do	1	2	3	4
Q30	A university education is more important for a boy than for a girl	1	2	3	4
Q31	On the whole, men make better business executives than women do	1	2	3	4
Q32	Being a housewife is just as fulfilling as working for pay	1	2	3	4

Original: 1 = strongly agree; 4 = strongly disagree

Lower original scores meant more traditional

Reverse Formula:

- reversed score = 5 – original score

After reverse-coding, the scale is easier to interpret, with higher scores representing more traditional gender-role attitudes.



Creating our Traditional Gender-role Attitudes Score

Reverse-Coding

Cronbach's Alpha

Traditional Gender Role
Attitude Score

- Q28–Q31 were **not** from an established traditional gender-role attitudes questionnaire.
- **Cronbach's alpha** was used to check whether the four items behaved **consistently** enough to be combined into one scale.
 - Does not prove construct validity, but it supports combining the items.
- Did the test before and after reverse-coding to confirm that reverse-coding only changed the direction of interpretation, not the relationships among the items.

For both before and after reverse-coding:

$$\alpha = 0.767$$

Showed acceptable internal consistency, so we combined it into one traditional gender-role attitude score



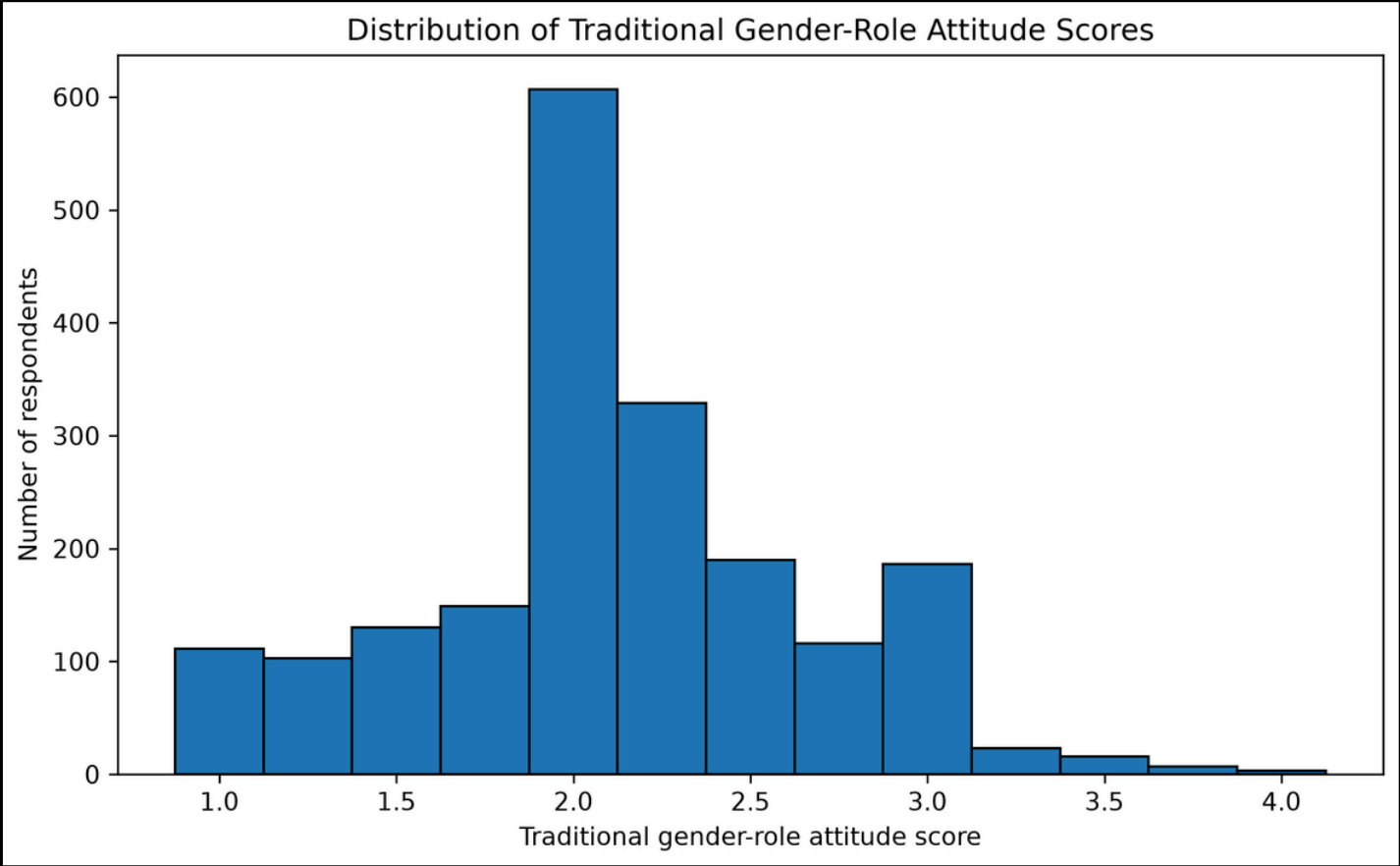
Creating our Traditional Gender-role Attitudes Score

Reverse-Coding

Cronbach's Alpha

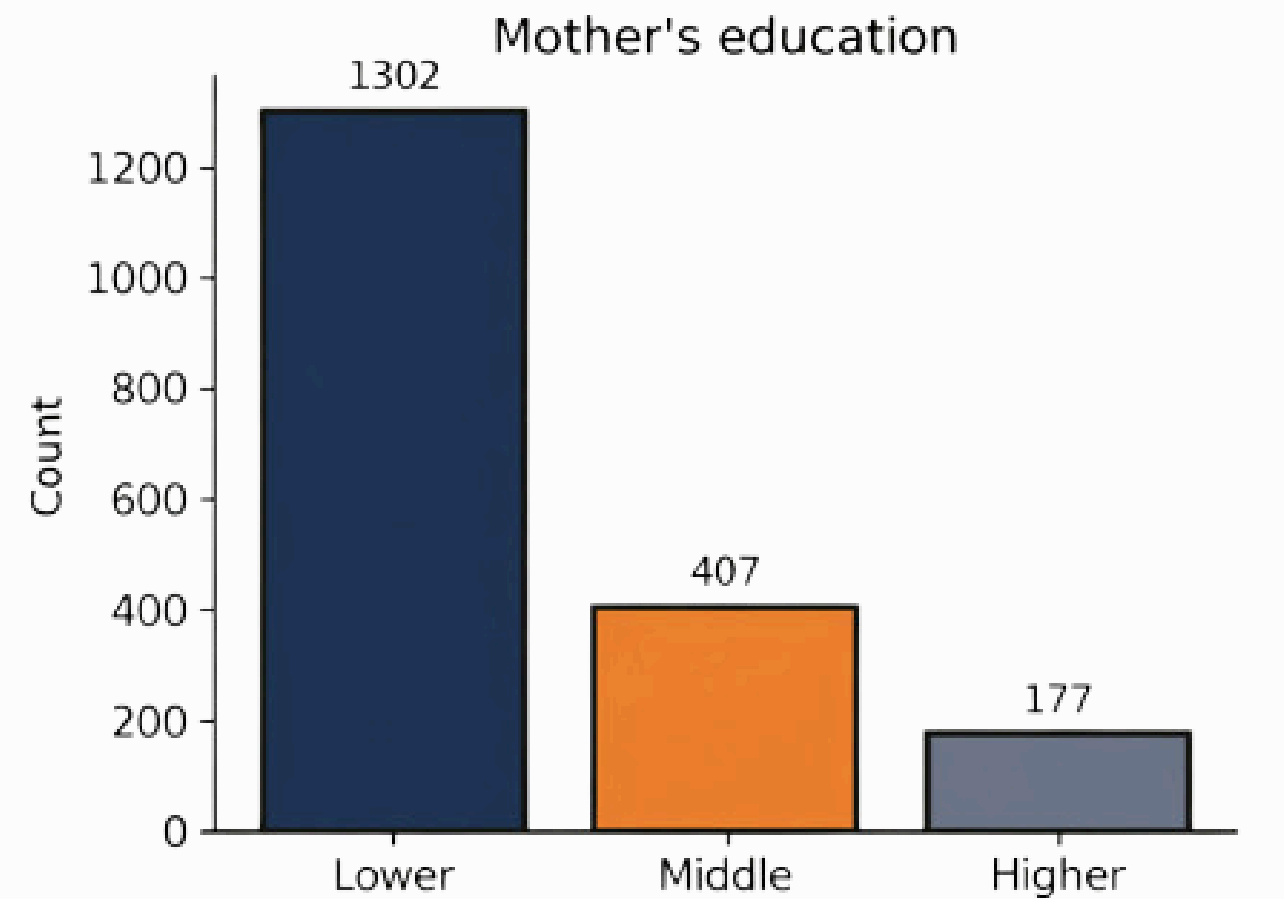
Traditional Gender Role
Attitude Score

- Composite score calculated as the mean of Q28_rev, Q29_rev, Q30_rev, and Q31_rev.
- Only respondents with valid responses to all four items were assigned a composite score.

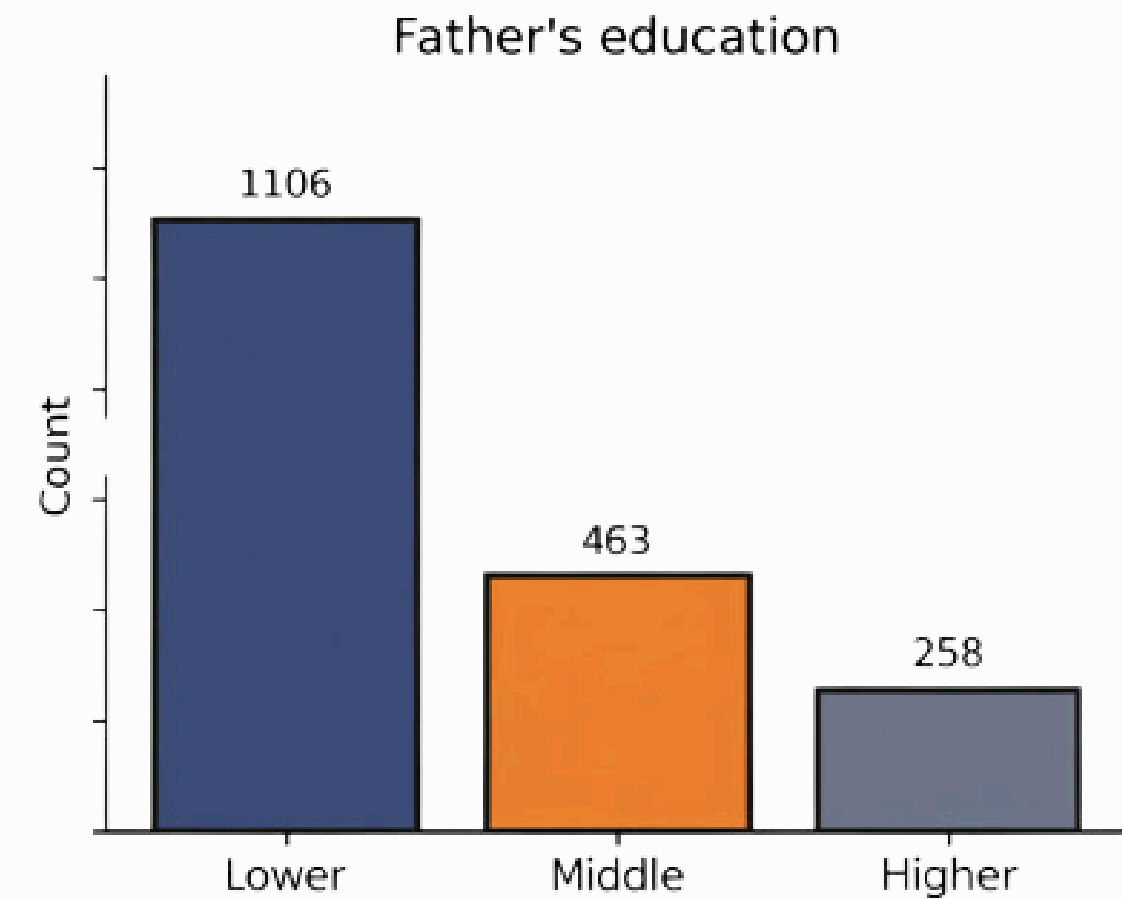


Valid N	1970
Mean	2.117
Standard deviation	0.556
Median	2
Minimum	1
Maximum	4
Skewness	0.101
Kurtosis	0.05

Descriptive Statistics

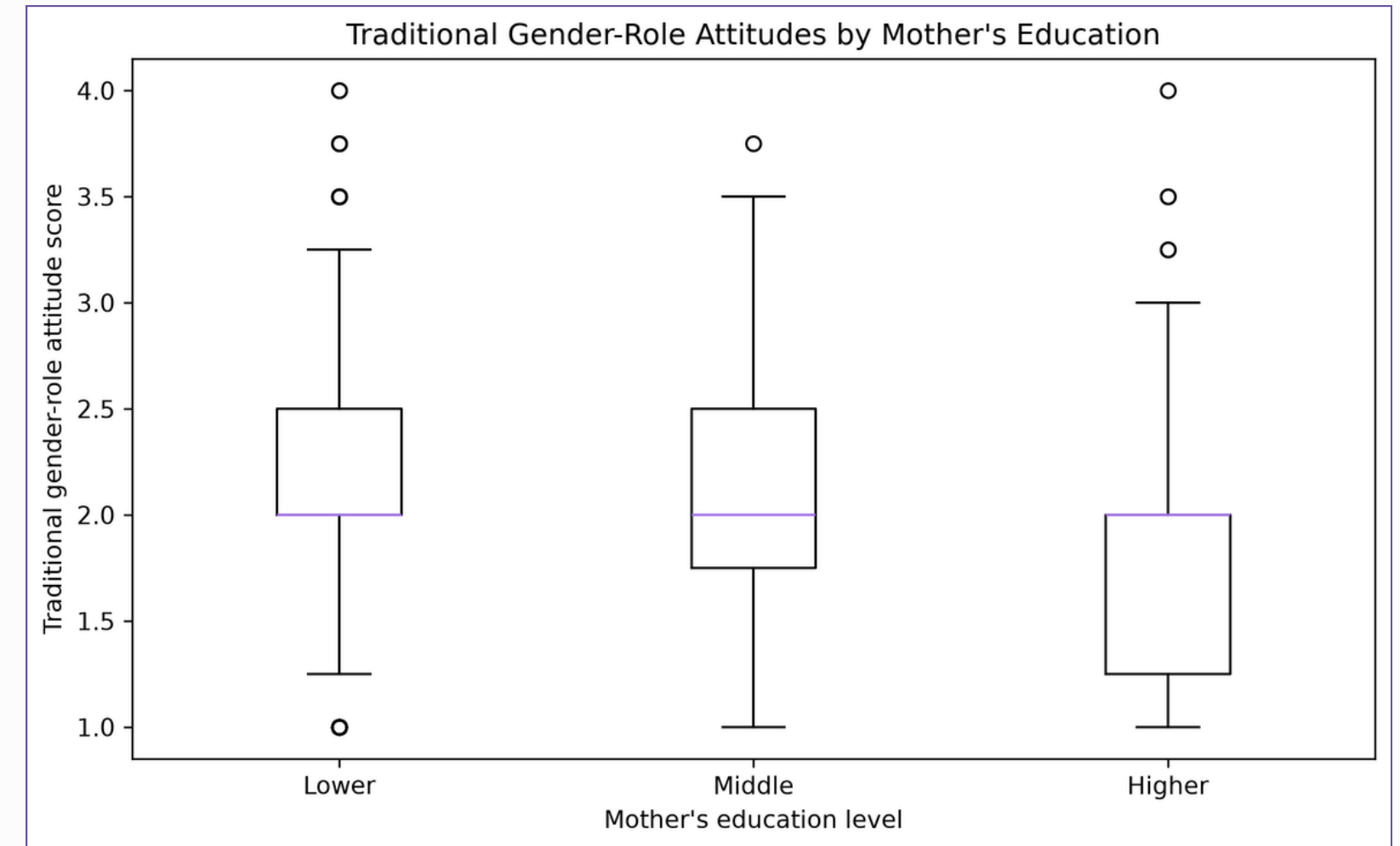
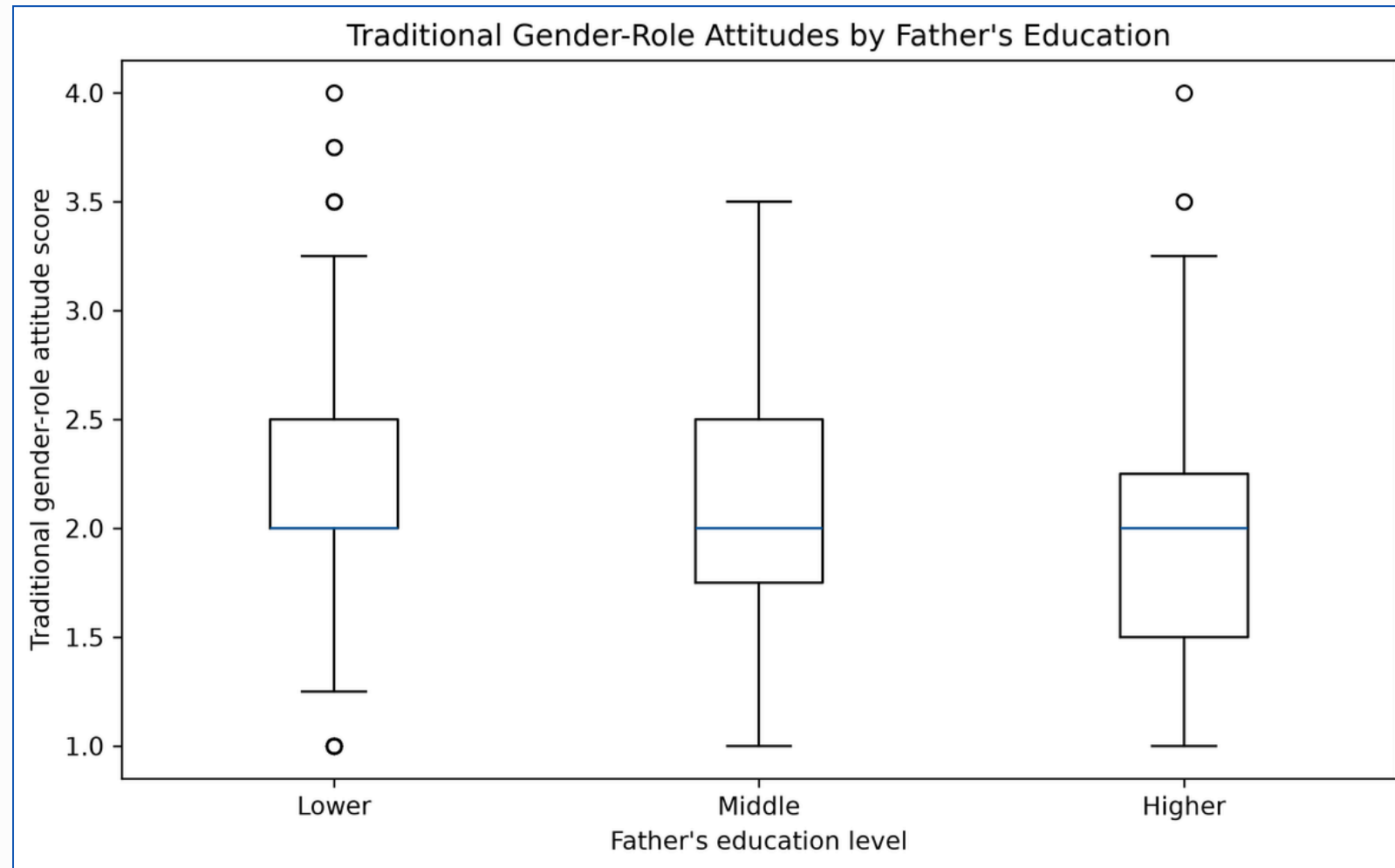


mother_education_label	count	mean	std	median
Lower	1279	2.1671227521501200	0.5242290798044370	2.0
Middle	397	2.054156171284640	0.5851392873922840	2.0
Higher	175	1.8371428571428600	0.6192949802622310	2.0



father_education_label	count	mean	std	median
Lower	1083	2.175207756232690	0.5218797145557000	2.0
Middle	456	2.06359649122807	0.5601818812845790	2.0
Higher	252	1.8988095238095200	0.6293601470143880	2.0

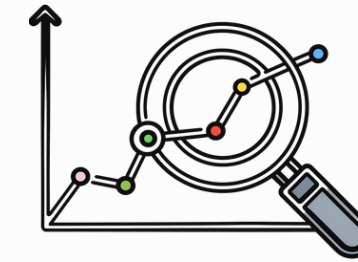
Initial Interpretation



For both maternal and paternal education, the overall distribution of traditional gender-role attitude scores shifts slightly downward as parental education increases.

Since lower scores represent less traditional attitudes, this suggests that higher parental education is associated with less traditional gender-role attitudes.

Spearman's R Correlation



parental_education_spearman_correlations

Parental education variable	N	Spearman rho	p-value
Mother's education	1851	-0.17064740458044000	1.45595590050982E-13
Father's education	1791	-0.16002425236529900	9.66061514092643E-12

Mother's Education

There was a statistically significant, weak negative correlation with traditional gender-role attitudes, $r = -.171$, $p < .001$.

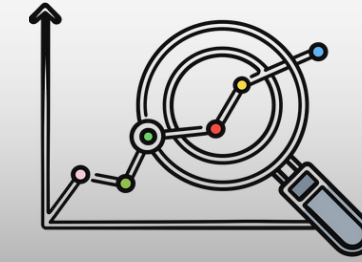
Higher maternal education was associated with less traditional gender-role attitudes..

Father's Education

Father's education: There was a statistically significant, weak negative correlation with traditional gender-role attitudes, $r = -.160$, $p < .001$.

Higher paternal education was associated with less traditional gender-role attitudes.

Spearman's R Correlation



Both parents, $p < .001$; therefore, we reject the null hypotheses and accept our alternative hypothesis:

H1a: Mother's education level is associated with traditional gender-role attitudes.

H1b: Father's education level is associated with traditional gender-role attitudes

What we have completed so far:

- Completed the literature review and established the rationale for the study
- Completed data cleaning and preparation
- Developed and validated the Traditional Gender-Role Attitudes composite score
- Completed the descriptive statistical analyses
- Completed the Spearman's rank correlation analyses addressing the main research question



What we will be working on:

- Explore potential moderating factors (e.g. religiosity, childcare responsibilities, & technology exposure)
- Develop a multivariable model examining how maternal and paternal education levels jointly predict Traditional Gender-Role Attitude scores
 - Extend the model by including other relevant factors to assess their additional contribution

**WORK IN
PROGRESS**

Thank You
for listening



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